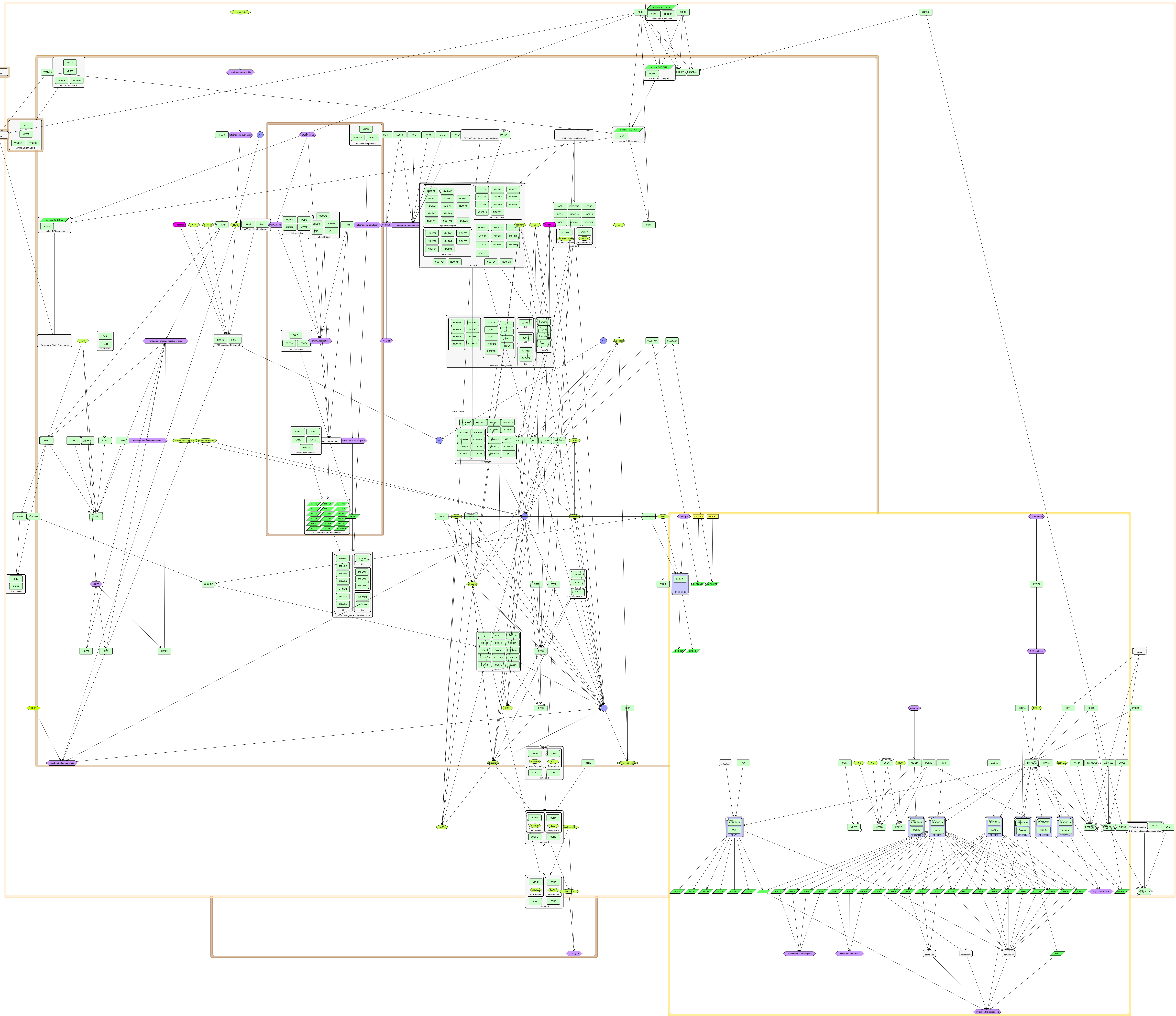
SCF-Cdc4 complex
FBXW7
SCF-Cdc4 ubiquitin ligase complex
UPS
NDN
PPARGC1A
T295
T299
Ub
AMPK
CAPN1
PROX1
PINK1
PRKN
UPS
ZNF746
ZNF746
Ub
nucleus
PARP1
DNA damage
PARP1
SIRT1
SNCA
ZNF746
PPARGC1A
GSK3B
NAD depletion
NAD(+)
PPARGC1A
Ac
PPARGC1A
S538
T177
MAPK p38
KAT2A
acetyl-CoA
MEF2D
autophagy
CDK5
MEF2D
ESRRA
YY1
GABPA
NRF1
PPARA
mTORC1
mitochondrial biogenesis
TF PPARA
TF ERRa
TF YY1
TF NRF2
TF NRF1
TF MEF2D
TF MEF2C
PPARA
ESRRA
YY1
GABPA
NRF1
MEF2D
MEF2C
fatty acid oxidation
SIRT3
UCP2
IDH3A
IDH3B
IDH3G
NDUFS8
ATP5MC1
COX5A
CYCS
COX4I1
COX5A
COX5B
COX6A1
COX6B1
COX6C
COX7A2
COX7B
COX7C
COX8A
CYCS
SDHB
SDHC
SDHD
ATP5F1B
ATP5F1C
complex IV
complex II
complex V
POLRMT
TFAM
TFB2M
TFB1M
mitochondrial transcription
SURF1
VDAC1
TOMM20
mitochondrial transport
mitochondrial biogenesis
RNS
NO
SNO
MEF2C
MEF2C
ROS
mutated A53T
SNCA
SOH
MEF2C
PPARGC1A
CHCHD2
TF CHCHD2
COX4I2
CHCHD2
hypoxia
ROS
CHCHD4
CHCHD2
SLC25A27
SLC25A14
SLC25A27
SLC25A14
ROS
PARK7
SLC25A27
SLC25A14
ROS
CCCP
peroxynitrite
SIRT3
Ac
NDUFA9
NDUFA10
PINK1
O2
superoxide
hydrogen peroxide
CHCHD2
SOD1
H+
mitochondrial depolarization
membrane permeability
superoxide
purine nucleotide
unsaturated fatty acid
Tiopronin
diazoxide
ATP
K+
NDUFA9
P
NDUFA10
NDUFA1
NDUFA2
NDUFA3
NDUFA4
NDUFA5
NDUFA6
NDUFA7
NDUFA8
NDUFA11
NDUFA12
NDUFA13
alpha subcomplex
NDUFB1
NDUFB2
NDUFB3
NDUFB4
NDUFB5
NDUFB6
NDUFB7
NDUFB8
NDUFB9
NDUFB10
NDUFB11
beta subcomplex
NDUFS1
NDUFS2
NDUFS3
NDUFS4
NDUFS5
NDUFS6
NDUFS7
NDUFS8
Fe-S protein
NDUFV1
NDUFV2
NDUFV3
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND4L
MT-ND5
MT-ND6
NDUFAB1
NDUFAF1
NDUFC1
NDUFC2
complex I
acetylated
SDHB
Ac
SDHA
Fe-S cluster
FAD
Iron sulfur protein
flavoprotein
SDHC
SDHD
complex II
ubiquinone
mutated A53T
H+
NADH
NAD(+)
rotenone
O2
superoxide
UQCRH
UQCRFS1P1
UQCRQ
iron-sulfur cluster
heme b
iron-sulfur subunit
MT-CYB:heme b
complex III
CHCHD2:CHCHD10:CYCS
CYCS
MT-CO1
MT-CO2
MT-CO3
COX7B
COX7C
COX8A
complex IV
SIRT5
myxothiazol
O2
H2O
ATP5PD
ATP5MG
ATP5PF
MT-ATP8
ATP5F1E
ATP5F1EP2
F(0)
F(1)
complex V
ADP
K+
H+
ROS
Ca2+
KCNJ8
KCNJ11
ATP-sensitive K+ channel
SLC25A27
SLC25A14
UCP2
UCP3
mitochondrial dysfunction
superoxide
MUL1
VPS35
VPS26A
VPS26B
VPS26:VPS35:MUL1
peroxisome
MUL1
VPS35
VPS26A
VPS26B
VPS26:VPS35:MUL1
SIRT3
SDHB
SDHA
Fe-S cluster
FAD
Fe-S protein
flavoprotein
SDHC
SDHD
complex II
OXPHOS assembly factors
SD2H/SC10E
PARK7
OXPHOS subunits encoded in mtDNA
OXPHOS assembly factors
CYCS
Ac
superoxide
O2
superoxide
ubiquinol
succinic acid
TCA cycle
ubiquinone
fumaric acid
SDHB
SDHA
Fe-S cluster
FADH2
Fe-S protein
flavoprotein
SDHC
SDHD
complex II
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND4L
MT-ND5
MT-ND6
CI
MT-CYB
CIII
MT-CO1
MT-CO2
MT-CO3
CIV
MT-ATP6
MT-ATP8
CV
OXPHOS subunits encoded in mtDNA
MT-TC
MT-TL1
MT-TS1
MT-TE
MT-TL2
MT-TS2
MT-TF
MT-TN
MT-TT
MT-TH
MT-TP
MT-TV
MT-TI
MT-TQ
MT-TW
MT-TK
MT-TR
MT-RNR1
mitochondrial tRNAs and rRNA
AARS2
DARS2
GARS
KARS
RARS2
Mt tRNA synthetases
POLG2
POLG
MTFMT
MTPAP
Mt replication
mtDNA damage
mtDNA replication
SUCLA2
DGUOK
RRM2B
TK2
SUCLG1
Mt-dNTP pool
mRNA
mitochondrial DNA
mtDNA repair
mitochondrial translation
mitochondrial transcription
TFAM
POLG
ERCC6
ERCC8
Mt-DNA repair
MRPL3
MRPS16
MRPS22
Mt ribosomal proteins
nucleoid
NDUFAF1
NDUFAF6
NDUFAF2
NDUFAF5
NDUFAF3
ACAD9
NDUFAF4
FOXRED1
CI
COX10
SCO1
COX14
SCO2
COX15
SURF1
FASTKD2
TACO1
LRPPRC
CIV
SDHAF1
CII
BCS1L
CIII
ATPAF2
TMEM70
CV
ABCB7
BOLA3
NUBPL
NFU1
Fe-S
OXPHOS assembly factors
mitochondrial matrix
CDK5
P
Y182
T180
MAPK14
P
MAPK12
mutated A141S
HTRA2
mutated G399S
HTRA2
P
S400
N-3
HTRA2
HTRA2
HTRA2
PARL
HAX1
HAX1:PARL
HAX1
TRAP1
ROS
PINK1
PINK1
mt UPR
HTRA2
P
TRAP1
LONP1
CLPB
HSPD1
HSPE1
HSPA9
HSPD1
HSPE1
HSPA9
CLPP
proteolysis
mt UPR
chaperone-mediated protein folding
chaperone-mediated protein folding
PINK1
PINK1
PRKN
PINK1:PRKN
PRKN
Ub
PINK
TFAM
mitochondrial respiratory chain
Respiratory Chain Components
ribosome
mitochondrion-attached ribosome
TOMM20
&
PINK1
ribosome
ribosome
nuclear RCC RNA
PUM1
nuclear RCC complex
TOMM20
nuclear RCC RNA
PINK1
nuclear RCC complex
PINK1
PUM1
&
PINK1
nuclear RCC RNA
PUM1
nuclear RCC complex
HNRNPF
&
PRKN
nuclear RCC RNA
PUM1
HNRNPF
nuclear RCC complex
mitochondrion
neuron


peroxynitrite
ribosome
TOMM20
MUL1
VPS35
VPS26A
VPS26B
VPS26:VPS35:MUL1
ribosome
MUL1
VPS35
VPS26A
VPS26B
VPS26:VPS35:MUL1
membrane permeability
TRAP1
mitochondrial dysfunction
Ca2+
mtDNA repair
MRPL3
MRPS16
MRPS22
Mt ribosomal proteins
CLPP
LONP1
HSPD1
HSPA9
CLPB
HSPE1
OXPHOS subunits encoded in mtDNA
C106
PARK7
OXPHOS assembly factors
PINK1
nuclear RCC RNA
PUM1
HNRNPF
nuclear RCC complex
PRKN
nuclear RCC RNA
PUM1
nuclear RCC complex
HNRNPF
Ub
ZNF746
ZNF746
nuclear RCC RNA
PUM1
nuclear RCC complex
nuclear RCC RNA
PINK1
nuclear RCC complex
diazoxide
ATP
Tiopronin
P
TRAP1
ROS
KCNJ8
KCNJ11
ATP-sensitive K+ channel
mtDNA damage
POLG2
POLG
MTFMT
MTPAP
Mt replication
SUCLA2
DGUOK
RRM2B
TK2
SUCLG1
Mt-dNTP pool
TFAM
mitochondrial translation
proteolysis
chaperone-mediated protein folding
NDUFA9
P
NDUFA10
NDUFA1
NDUFA2
NDUFA3
NDUFA4
NDUFA5
NDUFA6
NDUFA7
NDUFA8
NDUFA11
NDUFA12
NDUFA13
alpha subcomplex
NDUFB1
NDUFB2
NDUFB3
NDUFB4
NDUFB5
NDUFB6
NDUFB7
NDUFB8
NDUFB9
NDUFB10
NDUFB11
beta subcomplex
NDUFS1
NDUFS2
NDUFS3
NDUFS4
NDUFS5
NDUFS6
NDUFS7
NDUFS8
Fe-S protein
NDUFV1
NDUFV2
NDUFV3
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND4L
MT-ND5
MT-ND6
NDUFAB1
NDUFAF1
NDUFC1
NDUFC2
complex I
rotenone
O2
myxothiazol
UQCRH
UQCRFS1P1
UQCRQ
BCS1L
UQCR10
UQCR11
UQCRB
UQCRC1
UQCRC2
UQCRFS1
iron-sulfur cluster
iron-sulfur subunit
MT-CYB
heme b
MT-CYB:heme b
complex III
O2
PUM1
Respiratory Chain Components
ROS
PARL
HAX1
HAX1:PARL
chaperone-mediated protein folding
KCNJ8
KCNJ11
ATP-sensitive K+ channel
POLG
ERCC6
ERCC8
Mt-DNA repair
nucleoid
mtDNA replication
mt UPR
NDUFAF1
NDUFAF6
NDUFAF2
NDUFAF5
NDUFAF3
ACAD9
NDUFAF4
FOXRED1
CI
COX10
SCO1
COX14
SCO2
COX15
SURF1
FASTKD2
TACO1
LRPPRC
CIV
SDHAF1
CII
BCS1L
CIII
ATPAF2
TMEM70
CV
ABCB7
BOLA3
NUBPL
NFU1
Fe-S
OXPHOS assembly factors
K+
superoxide
SLC25A14
SLC25A27
mitochondrion
ATP5IF1
ATP5MC1
ATP5MC2
ATP5MC3
ATP5MF
ATP5PO
ATP5PB
ATP5MG
ATP5PD
ATP5MGL
ATP5ME
MT-ATP6
ATP5PF
MT-ATP8
F(0)
ATP5F1A
ATP5F1B
ATP5F1C
ATP5F1D
ATP5F1E
ATP5F1EP2
F(1)
complex V
UCP2
UCP3
SLC25A14
SLC25A27
ADP
K+
PINK1
MAPK12
Y180
T180
MAPK14
HTRA2
CDK5
mitochondrial respiratory chain
unsaturated fatty acid
purine nucleotide
AARS2
DARS2
GARS
KARS
RARS2
Mt tRNA synthetases
mitochondrial DNA
mitochondrial transcription
MT-TC
MT-TL1
MT-TS1
MT-TE
MT-TL2
MT-TS2
MT-TF
MT-TN
MT-TT
MT-TH
MT-TP
MT-TV
MT-TI
MT-TQ
MT-TW
MT-TK
MT-TR
MT-RNR1
mitochondrial tRNAs and rRNA
mRNA
PRKN
Ub
CHCHD4
P
S400
N
HTRA2
SNCA
NADH
mutated A53T
SNCA
H+
ATP
CHCHD2
ROS
hypoxia
SLC25A14
SLC25A27
DNA damage
PINK1
PRKN
PINK1:PRKN
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND4L
MT-ND5
MT-ND6
CI
MT-CYB
CIII
MT-CO1
MT-CO2
MT-CO3
CIV
MT-ATP6
MT-ATP8
CV
OXPHOS subunits encoded in mtDNA
CHCHD2
mt UPR
ubiquinol
SIRT5
Ac
CYCS
GHITM
CHCHD2
reduced
CYCS
CHCHD2:GHITM:CYCS
PARK7
CHCHD2
TF CHCHD2
SLC25A14
SLC25A27
PARP1
MT-CO1
MT-CO2
MT-CO3
COX4I1
COX4I2
COX5A
COX5B
COX6A1
COX6A2
COX7A1
COX7A2L
COX7A2
COX7B
COX7C
COX8A
complex IV
reduced
CYCS
CHCHD2
COX4I2
NAD depletion
AMPK
HSPA9
HSPD1
HSPE1
H2O
oxidized
CYCS
H+
SOD1
autophagy
ESRRA
NAD(+)
SIRT1
SNCA
PROX1
CCCP
acetylated
SDHB
SDHA
Fe-S cluster
Iron sulfur protein
Ac
FAD
flavoprotein
SDHC
SDHD
complex II
mitochondrial depolarization
ubiquinone
SIRT3
hydrogen peroxide
mTORC1
YY1
CDK5
RNS
NO
mutated A53T
SNCA
ROS
MEF2D
MEF2C
NRF1
GABPA
PPARGC1A
S538
T177
P
PPARA
acetyl-CoA
KAT2A
PPARGC1A
Ac
MAPK p38
GSK3B
SDHB
SDHA
Fe-S cluster
FAD
Fe-S protein
flavoprotein
SDHC
SDHD
complex II
succinic acid
NAD(+)
P
T295
PPARGC1A
YY1
TF YY1
MEF2D
P
MEF2C
SNO
MEF2C
SOH
PPARGC1A
MEF2D
TF MEF2D
PPARGC1A
NRF1
TF NRF1
PPARGC1A
GABPA
TF NRF2
PPARGC1A
ESRRA
TF ERRa
PPARGC1A
MEF2C
TF MEF2C
PPARGC1A
PPARA
TF PPARA
PPARGC1A
Ac
S570
T177
PPARGC1A
Ac
T262
T299
ZNF746
SCF-Cdc4 complex
FBXW7
SCF-Cdc4 ubiquitin ligase complex
NDN
SDHB
SDHA
Fe-S cluster
FADH2
Fe-S protein
flavoprotein
fumaric acid
SDHC
SDHD
complex II
UCP2
IDH3A
IDH3G
NDUFS8
ATP5MC1
IDH3B
CYCS
TFB1M
TFB2M
TFAM
POLRMT
VDAC1
SURF1
TOMM20
ATP5F1C
COX5A
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SDHC
COX6C
ATP5F1B
COX8A
COX5B
COX7B
COX7C
COX7A2
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COX6B1
COX6A1
fatty acid oxidation
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mitochondrial transport
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complex V
complex IV
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